

EX. NO: 1 IMPLEMENTATION OF DATA ENCRYPTION STANDARD (DES)

AIM

To implement the Data Encryption Standard (DES) encryption and decryption algorithm using Java.

DESCRIPTION

Data Encryption Standard (DES) is a symmetric key block cipher that encrypts data in 64-bit blocks using a 56-bit secret key. It performs a sequence of permutations, substitutions, and XOR operations through 16 rounds of processing. The same key is used for both encryption and decryption.

ALGORITHM

STEP-1: Read the plain text message from the user.

STEP-2: Generate a DES secret key.

STEP-3: Initialize the DES cipher in encryption mode.

STEP-4: Encrypt the plain text.

STEP-5: Display the encrypted text.

STEP-6: Initialize the cipher in decryption mode.

STEP-7: Decrypt the encrypted text.

STEP-8: Display the original message.

CODE

```
import javax.crypto.Cipher;
import javax.crypto.KeyGenerator;
import javax.crypto.SecretKey;
import java.util.Base64;
import java.util.Scanner;

public class DESExample {

    public static void main(String[] args) {
        try {

            Scanner sc = new Scanner(System.in);
```

```
System.out.print("Enter the message: ");

String message = sc.nextLine();


// Generate DES Key

KeyGenerator keyGenerator = KeyGenerator.getInstance("DES");
SecretKey secretKey = keyGenerator.generateKey();


// Encryption

Cipher cipher = Cipher.getInstance("DES");
cipher.init(Cipher.ENCRYPT_MODE, secretKey);

byte[] encryptedBytes = cipher.doFinal(message.getBytes());

String encryptedText = Base64.getEncoder().encodeToString(encryptedBytes);

System.out.println("\nEncrypted Message:");
System.out.println(encryptedText);


// Decryption

cipher.init(Cipher.DECRYPT_MODE, secretKey);

byte[] decryptedBytes = cipher.doFinal(encryptedBytes);

String decryptedText = new String(decryptedBytes);

System.out.println("\nDecrypted Message:");
```

```
        System.out.println(decryptedText);

    } catch (Exception e) {
        e.printStackTrace();
    }
}
}
```

Output

Clear

Enter the message: run

Encrypted Message:
HG8N2LO3uGc=

Decrypted Message:
run

=== Code Execution Successful ===

RESULT

Thus, the Data Encryption Standard (DES) encryption and decryption algorithm was implemented successfully using Java.